TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Testing of Flame-retardant, Water Repellent and Crease-resistant Fabrics

Standard methods applied to the Experiment 1–3 fabrics — flame, spray and recovery numbers that a buyer, not just a bench, will accept.

WP-4202 · TEXTILE FINISHING SESSIONAL · EXPERIMENT 4 · SPECIALIZATION — WET PROCESSING

1. Aim and theory

Aim: to test, by recognised standard methods, the fabrics produced in Experiments 1–3: (i) **vertical flame test** and **limiting oxygen index (LOI)** concept for the flame-retardant cotton of Experiment 1B; (ii) **spray test (ISO 4920)**, with a note on the Bundesmann rain test, for the water-repellent fabrics of Experiment 1A; (iii) **crease-recovery angle (AATCC 66 principle)** and **durable-press appearance rating (AATCC 124)** for the crease-resistant cotton of Experiment 3; and (iv), as an additional standard test, the quantitative **antibacterial assessment (AATCC 100)** for the antibacterial fabric type of Experiment 2D — with a fully worked % reduction calculation. Each result is interpreted against its untreated control, because a finish is only ever proved by a difference.

Why standard methods matter. Experiments 1–3 used screening checks: beading, a demonstration burn, a comparative fold test. Those show a finish is present. A standard method fixes every variable — specimen size, conditioning, flame height, spray volume, creasing load, recovery time, rating scale and the number of replicates — so that two laboratories testing the same fabric reach the same verdict, and a specification ("spray rating ≥ 4; char length ≤ 100 mm; W + F ≥ 270°") can be written into a purchase contract. In this experiment the procedure *is* the lesson: change one uncontrolled variable and you have changed the result, not the fabric.

Two general rules govern all four sections. First, **conditioning**: specimens are conditioned at 20 ± 2 °C and 65 ± 4% relative humidity (flame specimens per their own standard's drying/conditioning clause) and tested promptly, because moisture content moves every one of these properties. Second, **replication**: flammability and recovery vary across a piece; standards require multiple specimens (typically five) in warp and, where relevant, weft directions, and the report gives individual values plus the mean — never a single heroic specimen.

2. Flame-retardant testing

2.1 Vertical flame test

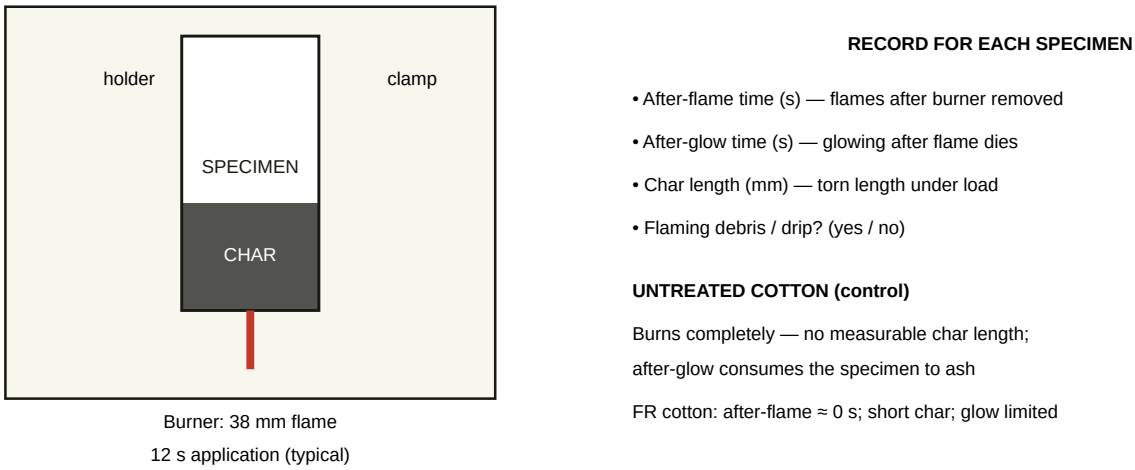


Fig. 1 — Vertical flame test apparatus: specimen held vertically, standard flame applied to the bottom edge, after-flame, after-glow and char length recorded.

Principle and procedure (ASTM D6413 / ISO 15025 family). A conditioned specimen (commonly 300 mm × 76 mm) is held vertically in a holder inside a draught-free cabinet. A standard burner flame (about 38 mm high) is applied to the centre of the bottom edge for a fixed time (12 s in ASTM D6413). On removal, three quantities are timed/measured: **after-flame time** (how long the specimen itself keeps flaming), **after-glow time** (how long glowing combustion continues after the flame is out — critical for cotton, which glows), and **char length** (the length of damaged specimen, measured by hanging a specified tearing load on the char and measuring the tear). Whether flaming debris falls is also recorded.

QUANTITY	UNTREATED COTTON (TYPICAL)	FR COTTON, EXP. 1B (TYPICAL PASS BEHAVIOUR)
After-flame time	Burns until consumed	0–2 s (self-extinguishing)
After-glow time	Long; glow spreads to ash	Short; glow confined to char zone
Char length	Full length (specimen destroyed)	Commonly 60–100 mm; many specifications require ≤ 100–150 mm — always quote the specification actually used

Interpretation. A durable FR finish is proved when after-flame is essentially zero and char length is short *and* the same result is obtained after the specified number of launderings (durability is part of the claim from Experiment 1). A specimen that passes before washing and fails after five washes was a deposit, not a finish.

2.2 Limiting oxygen index (LOI)

LOI (ISO 4589 / ASTM D2863) is the minimum oxygen concentration, in an oxygen–nitrogen mixture flowing past a small vertically held specimen, that just sustains candle-like burning. It is a material index from a dedicated apparatus — in this sessional it is taught as a concept with reference values, not performed on the bench rig.

$$\text{LOI (\%)} = \text{minimum } [O_2] \text{ that sustains burning} \cdot \text{Air} = 21\%\ O_2$$

MATERIAL	LOI (TYPICAL)	READING
Untreated cotton	≈ 18.4%	Below air — burns readily
FR-treated cotton (Exp. 1B type)	≈ 27–30%	Above air — tends to self-extinguish in air
Polyester (for comparison)	≈ 20–21%	Borderline; melts and drips, a different hazard

Caution: LOI is measured on a small specimen in a controlled gas stream. A high LOI supports an FR claim; it does not by itself certify garment safety, which depends on construction, fit and the full test standard for the end use (e.g. protective-clothing standards). Never convert an LOI value into “fireproof”.

3. Water-repellent testing — spray test and Bundesmann

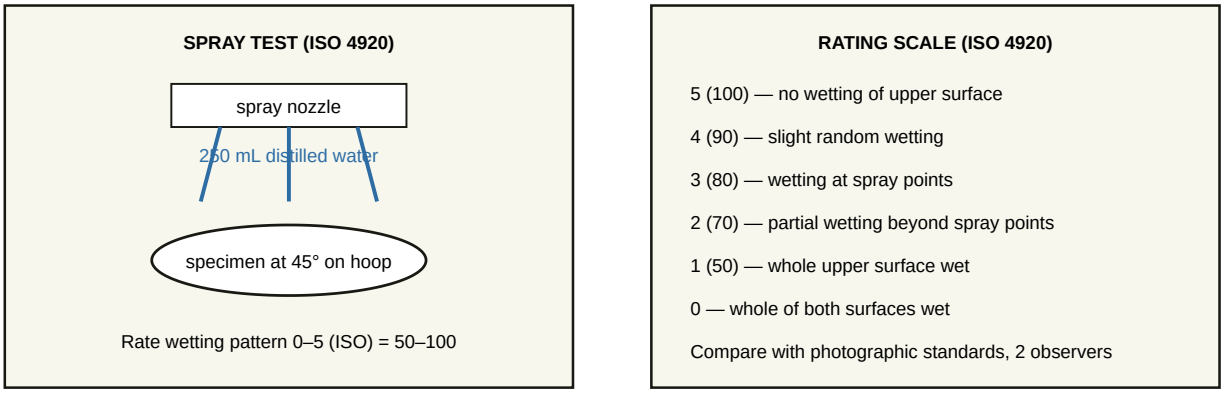


Fig. 2 — Spray test apparatus (left) and the ISO 4920 rating scale (right). The AATCC 22 numbers (50–100) for the same patterns are shown in brackets in the text table.

Procedure (ISO 4920 / AATCC 22). A 180 mm × 180 mm conditioned specimen is mounted, face up, on a hoop held at 45°. Distilled water (250 mL at 27 ± 1 °C) is poured through the standard spray nozzle from a fixed height so it showers the specimen centre over 25–30 s. Excess water is knocked off with two sharp taps, and the wetting pattern is rated immediately against the photographic standard scale by comparison of how much of the surface is wetted:

ISO GRADE	AATCC RATING	APPEARANCE
5	100	No sticking or wetting of the upper surface
4	90	Slight random sticking or wetting
3	80	Wetting at the spray points
2	70	Partial wetting beyond the spray points
1	50	Complete wetting of the whole upper surface
0	0	Complete wetting of both surfaces (strike-through)

Typical interpretation for Experiment 1A fabrics. Untreated cotton rates 0 — it wets completely at once. A correctly cured fluorocarbon finish at 19.5 g/kg add-on typically rates ISO 4–5 (AATCC 90–100) when new, falling after laundering unless the finish is durable; a one-grade fall after the specified washes is a common acceptance limit. Polyester starts higher untreated (its hydrophobicity gives a deceptive ISO 2–3) — which is why the control must always be reported alongside.

Bundesmann note. The Bundesmann test (ISO 9865) goes further: specimens are exposed to a standardised rain shower for 10 min while being rubbed from behind, and both the visual rating and the *volume of water penetrated* (and absorbed) are measured. Use it when the claim concerns rainwear performance under prolonged rain, not just surface beading; the spray test rates the surface, the Bundesmann also catches penetration.

4. Crease-resistance testing

4.1 Crease-recovery angle — AATCC 66 principle

A rectangular specimen (40 mm × 15 mm) is folded face-to-face (or back-to-back), compressed under a standard load (typically 500 g or 1 kg depending on the instrument) for a fixed time (commonly 5 min), then transferred to the recovery tester, released, and allowed to recover for a fixed time (commonly 5 min) before the angle between the two limbs is read from the protractor scale. Warp-folded and weft-folded specimens (fold across warp / across weft) are reported as the W + F sum. Dry recovery is standard; wet recovery (specimens soaked before folding) is the harsher variant that exposes cotton's wet creasing.

SPECIMEN (EXP. 3)	W + F RECOVERY ANGLE (TYPICAL RECORDED)	INTERPRETATION
Untreated cotton	≈ 156°	Poor — about 87° per direction on average

DMDHEU-treated, 3.5% owf	≈ 288°	Good apparel level; gain of 132° over control

$$\text{Gain} = 288° - 156° = 132° \cdot \text{Maximum possible sum} = 180° + 180° = 360°$$

Always state load, fold time and recovery time with the angle — a "288°" obtained with a longer recovery time is not the same result as the standard one, and examiners (and buyers) ask.

4.2 Appearance after washing — DP rating, AATCC 124

The durable-press (DP) rating judges what the customer actually sees. Treated fabric is washed and tumble-dried under standard conditions (repeatedly, typically assessed after 5 cycles), then three trained observers compare it, in a standard viewing cabinet, with photographic DP replicas and assign:

DP RATING	APPEARANCE
5	Smooth, pressed, original appearance
4	Slightly creased; acceptable for most apparel
3.5	Common commercial minimum for "easy-care" claims (state the buyer's limit used)
3	Noticeably creased; needs touch-up ironing
2	Badly creased
1	Severely creased, as untreated cotton after washing

The two crease tests answer different questions and both are needed: AATCC 66 measures an *instrumental fibre property* on a small strip; AATCC 124 measures *garment-relevant appearance* after real washing. A fabric can show a good recovery angle yet a mediocre DP rating if seams, construction or uneven finishing spoil its appearance — report both numbers for the Experiment 3 fabric and say so.

5. Additional standard test — antibacterial activity, AATCC 100

AATCC 100 is the quantitative counterpart of the zone test noted in Experiment 2D. Treated and untreated (control) swatches are inoculated with a known number of bacteria (a standard organism such as *Staphylococcus aureus* or *Klebsiella pneumoniae*), incubated in contact for 24 h, then shaken out in neutralising broth and counted by plate dilution. The result is the percent reduction against the control recovered after the same contact time:

$$\%\ \text{reduction} = [(A - B) \div A] \times 100$$

where A = colonies recovered from the untreated control after 24 h contact, and B = colonies recovered from the treated fabric after 24 h contact.

Worked example (use exactly these figures in the record)

QUANTITY	VALUE
Control recovery after 24 h, A	1.2×10^5 CFU
Treated recovery after 24 h, B	3.0×10^2 CFU

$$\% \text{ reduction} = [(120{,}000 - 300) \div 120{,}000] \times 100 = (119{,}700 \div 120{,}000) \times 100 = 99.75\%$$

Expressed as a log reduction: $\log_{10}(A) - \log_{10}(B) = \log_{10}(1.2 \times 10^5) - \log_{10}(3.0 \times 10^2) \approx 5.08 - 2.48 \approx$ **2.6 log reduction**. Both statements describe the same plate counts; buyers' specifications variously demand ≥ 99% reduction or ≥ 2 log — always report the raw A and B with the derived figure so either form can be checked. Note also what AATCC 100 does *not* show: it does not distinguish leaching from bonded action (that was the zone test's job, Exp. 2D), and a pass against one organism does not prove activity against all organisms.

> **READING A TEST REPORT — FIVE CHECKS**
>
> (1) Is the control reported? (2) Are conditions (load, time, temperature, washes) the standard ones? (3) How many specimens, and is it a mean? (4) Before or after washing — and how many washes? (5) Against which specification limit is it being judged? A number failing any of these checks is a bench observation, not a test result.

6. Consolidated results (Experiments 1–3 fabrics)

FINISH	TEST	CONTROL	TREATED (TYPICAL)	VERDICT AT TEACHING-LAB LEVEL
Flame retardant	Vertical flame	Consumed, full char	After-flame ≈ 0 s; char 60–100 mm	Effective; confirm after washing for durability
Flame retardant	LOI (reference)	≈ 18.4%	≈ 27–30%	Consistent with vertical-flame pass
Water repellent	Spray, ISO 4920	0	4–5 (AATCC 90–100)	Effective when cured; re-test after washes
Crease resistant	Recovery angle, W + F	≈ 156°	≈ 288° (gain 132°)	Good apparel recovery; report strength loss with it

Crease resistant	DP rating, AATCC 124	≈ 1–2	≈ 3.5–4 (target)	Meets common easy-care minimum if ≥ 3.5
Antibacterial	AATCC 100	A = 1.2×10^5 CFU	B = 3.0×10^2 CFU; 99.75% (≈2.6 log)	Strong antibacterial activity in this worked example

7. Precautions

- Flame cabinet draught-free, burner flame height checked before every set; a taller flame invents failures, a shorter one invents passes.
- Char length is measured with the standard tearing load specified by the method — pulling by hand gives whatever answer the hand chooses.
- Spray water at the specified temperature and volume, specimen tapped exactly as the standard directs; rating by two observers against the photographic standards, not from memory.
- Recovery-angle specimens cut precisely to size with warp/weft identified; a skewed specimen reads a false angle.
- Microbiology (AATCC 100) only under the laboratory's microbiology supervision, with sterilisation and disposal by its procedure. The worked example above is calculated from given plate counts; students do not culture pathogens unsupervised.
- Report every result with its control, its conditions and its number of specimens — see the five checks in Section 5.

8. Practice (viva preparation)

1. In the vertical flame test, why are after-flame, after-glow and char length all recorded? Which of the three matters most specifically for cotton, and why?
2. A fabric has LOI 24%. Will it burn in air? Answer carefully, in one or two sentences.
3. Convert an ISO spray grade of 4 to its AATCC rating and describe the surface appearance.
4. Why can untreated polyester score a misleadingly good spray rating, and how does the report protect against the misreading?
5. A treated fabric gives W + F = 270° and a control gives 150°. Calculate the gain and the treated fabric's recovery as a percentage of the theoretical maximum sum.
6. Using the AATCC 100 formula, find the % reduction if $A = 2.0 \times 10^6$ and $B = 2.0 \times 10^4$.
7. Why are both AATCC 66 and AATCC 124 needed to judge a crease-resistant fabric?

Answer key

1. After-flame shows flaming combustion continuing without the burner; after-glow shows smouldering/glowing combustion continuing after flames die; char length measures how far damage travelled. After-glow matters most for cotton because cellulosics characteristically glow and can be consumed (or re-ignite material nearby) long after the flame is out.
2. Its LOI (24%) is above air's 21%, so in the LOI apparatus it would not sustain burning in normal air and would tend to self-extinguish — but LOI is a small-specimen laboratory index, so this supports, and does not by itself guarantee, safe behaviour in a real fire.
3. AATCC 90. Slight random sticking or wetting of the upper surface, otherwise unwetted.
4. Polyester is intrinsically hydrophobic, so water beads on it even unfinished. The report protects against misreading by always publishing the untreated control rating beside the treated rating and by judging the finish on the improvement and its retention after washing.
5. Gain = 270 − 150 = 120°. As a percentage of the maximum sum: 270 ÷ 360 × 100 = 75%.
6. % reduction = [(2,000,000 − 20,000) ÷ 2,000,000] × 100 = 99.0% (a 2.0 log reduction).
7. AATCC 66 is an instrumental strip test of fibre recovery; AATCC 124 rates whole-fabric appearance after standard washing and drying, which is what the wearer experiences and what easy-care claims are sold on. Construction and uneven finishing can spoil the second while the first still looks good, so a complete judgement needs both.

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